Assignment 2

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Please note that

- **Released date: January 30, Tuesday.**
- **Due date: February 29, Thursday, by 11:55pm.**
- Late submission is **NOT** accepted.
- Please submit your answers as a PDF file with a name like "123010XXX ASS2.pdf" (Your student ID + ASS No.). You may either typeset you answers directly using computers, or scan your handwritten answers. (We recommend you use the printers on campus to scan. If you use your smartphone to scan, please limit the file size 10MB.)

Question 1. Label the following statements as true or false:

(a) An $n \times n$ matrix A is nonsingular if and only if the reduced row echelon form of A is I_n.

(b) If A and B are invertible, then $A + B$ is also invertible.

(c) If $A = A^{-1}$, then A must be equal to either I or $-I$.

(d) The transpose of an elementary matrix is an elementary matrix.

Solution

(a) True
(b) False (e.g. take $A = I$, $B = -I$)
(c) False (e.g. take $A = \text{diag}(1, -1)$)
(d) True

Question 2. Let A be an $m \times n$ matrix with the property that $A\mathbf{x} = \mathbf{0}$ for all $\mathbf{x} \in \mathbb{R}^n$. Show that $A = O$.

Solution

Define the vectors $\mathbf{e}_j \in \mathbb{R}^n$ by

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad \cdots, \quad \mathbf{e}_n = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}.$$

Method 1. We can write $I_n = [\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n]$. Then, by the property of the identity matrix and the definition of matrix-matrix multiplication, we have

$$\begin{aligned} A = AI_n &= A \begin{bmatrix} \mathbf{e}_1 & \mathbf{e}_2 & \cdots & \mathbf{e}_n \end{bmatrix} \\ &= \begin{bmatrix} A\mathbf{e}_1 & A\mathbf{e}_2 & \cdots & A\mathbf{e}_n \end{bmatrix} = \begin{bmatrix} \mathbf{0} & \mathbf{0} & \cdots & \mathbf{0} \end{bmatrix} = O, \end{aligned}$$

since $A\mathbf{e}_j = \mathbf{0}$ for all j by the given assumption.

Method 2. By the definition of matrix-vector multiplication, the j-th column of A is

$$\mathbf{a}_j = A\mathbf{e}_j = \mathbf{0}.$$

This is true for all $j \in \{1, \ldots, n\}$, so all columns of A are $\mathbf{0}$. Hence, $A = O$.

Question 3. Let A and B be $n \times n$ matrices. Show that if $AB = A$ and $B \neq I$, then A must be singular.

Solution

Suppose A^{-1} exists. Then multiplying both sides of $AB = A$ by A^{-1} from the left yields $A^{-1}AB = A^{-1}A$, which simplifies to $B = I$, which contradicts with the assumption that $B \neq I$. Hence, A must be singular.

Question 4. Let A be an $n \times n$ matrix. Show that if $A^{k+1} = O$ for some integer $k \geq 1$, then $I - A$ is nonsingular and $(I - A)^{-1} = I + A + A^2 + \cdots + A^k$.

Solution

Since $A^{k+1} = O$, we have

$$(I - A)(I + A + A^2 + \cdots + A^k) = I - A^{k+1} = I$$

and

$$(I + A + A^2 + \cdots + A^k)(I - A) = I - A^{k+1} = I.$$

Hence, by definition, $I-A$ is nonsingular and its inverse is $I+A+A^2+\cdots+A^k$.

Question 5. Let

$$A=\begin{bmatrix}1&2&4\\2&1&3\\1&0&2\end{bmatrix},\quad B=\begin{bmatrix}1&2&4\\2&1&3\\2&2&6\end{bmatrix},\quad C=\begin{bmatrix}1&2&4\\0&-1&-3\\2&2&6\end{bmatrix}.$$

(a) Find an elementary matrix E such that $EA=B$.

(b) Find an elementary matrix F such that $FB=C$.

(c) Is C row equivalent to A? Explain

Solution

(a) $E=\begin{bmatrix}1&0&0\\0&1&0\\1&0&1\end{bmatrix}$

(b) $F=\begin{bmatrix}1&0&0\\0&1&-1\\0&0&1\end{bmatrix}$

(c) Yes, because $C=FEA$, where E,F are elementary matrices.

Question 6. Compute the LU factorization of each of the following matrices.

$$\text{(a)}\begin{bmatrix}3&1\\9&5\end{bmatrix}\qquad\text{(b)}\begin{bmatrix}2&4\\-2&1\end{bmatrix}\qquad\text{(c)}\begin{bmatrix}-2&1&2\\4&1&-2\\-6&-3&4\end{bmatrix}$$

Solution

(a)

$$\begin{bmatrix}3&1\\9&5\end{bmatrix}\xrightarrow{R_2\to -3R_1+R_2}\begin{bmatrix}3&1\\0&2\end{bmatrix}$$

$$L=\begin{bmatrix}1&0\\3&1\end{bmatrix},\ U=\begin{bmatrix}3&1\\0&2\end{bmatrix}$$

(b)

$$\begin{bmatrix} 2 & 4 \\ -2 & 1 \end{bmatrix} \xrightarrow{R_2 \to R_1 + R_2} \begin{bmatrix} 2 & 4 \\ 0 & 5 \end{bmatrix}$$

$$L = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}, U = \begin{bmatrix} 2 & 4 \\ 0 & 5 \end{bmatrix}$$

(c)

$$\begin{bmatrix} -2 & 1 & 2 \\ 4 & 1 & -2 \\ -6 & -3 & 4 \end{bmatrix} \xrightarrow[R_3 \to -3R_1 + R_3]{R_2 \to 2R_1 + R_2} \begin{bmatrix} -2 & 1 & 2 \\ 0 & 3 & 2 \\ 0 & -6 & -2 \end{bmatrix} \xrightarrow{R_3 \to 2R_2 + R_3} \begin{bmatrix} -2 & 1 & 2 \\ 0 & 3 & 2 \\ 0 & 0 & 2 \end{bmatrix}$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 3 & -2 & 1 \end{bmatrix}, U = \begin{bmatrix} -2 & 1 & 2 \\ 0 & 3 & 2 \\ 0 & 0 & 2 \end{bmatrix}$$

Question 7. Find the LDU decomposition of the matrix

$$\begin{bmatrix} 1 & 1 & 1 \\ 3 & 5 & 6 \\ -2 & 2 & 7 \end{bmatrix}.$$

Solution

We perform row operations as follows:

$$\begin{bmatrix} 1 & 1 & 1 \\ 3 & 5 & 6 \\ -2 & 2 & 7 \end{bmatrix} \xrightarrow[R_3 \to 2R_1 + R_3]{R_2 \to -3R_1 + R_2} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 3 \\ 0 & 4 & 9 \end{bmatrix} \xrightarrow{R_3 \to -2R_2 + R_3} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 3 \\ 0 & 0 & 3 \end{bmatrix}$$

Therefore,

$$\begin{bmatrix} 1 & 1 & 1 \\ 3 & 5 & 6 \\ -2 & 2 & 7 \end{bmatrix} = LDU,$$

where

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ -2 & 2 & 1 \end{bmatrix}, D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}, U = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & \frac{3}{2} \\ 0 & 0 & 1 \end{bmatrix}.$$

Question 8. Let A be a 3×3 matrix and suppose that

$$2\mathbf{a}_1 + \mathbf{a}_2 - 4\mathbf{a}_3 = \mathbf{0},$$

where $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$ are the column vectors of A. Does the linear system $A\mathbf{x} = \mathbf{0}$ have a non-trivial solution? Is A invertible?

Solution

Let

$$\mathbf{x}_0 = \begin{bmatrix} 2 \\ 1 \\ -4 \end{bmatrix}.$$

By Definition II of matrix-vector multipliaction,

$$A\mathbf{x}_0 = 2\mathbf{a}_1 + \mathbf{a}_2 - 4\mathbf{a}_3 = \mathbf{0}.$$

Hence, $A\mathbf{x} = \mathbf{0}$ has a non-trivial solution $\mathbf{x} = \mathbf{x}_0$.
If A were invertible, then $A\mathbf{x} = \mathbf{0}$ has a unique solution which is $A^{-1}\mathbf{0} = \mathbf{0}$, the trivial solution. This is a contradiction. Therefore, A is not invertible.

Question 9. Let U be an $n \times n$ upper triangular matrix with non-zero diagonal entries. Prove that U is invertible and U^{-1} is also an upper triangular matrix.

Solution

We observe that it is possible to reduce the augmented matrix $[U|I_n]$ to $[I_n|V]$ as follows. First, we perform the type II row operations

$$R_1 \to \frac{1}{u_{11}} R_1,$$
$$\vdots$$
$$R_n \to \frac{1}{u_{nn}} R_n,$$

to reduce all diagonal entries of U to 1, which is possible since U has non-zero diagonal entries. The corresponding type II elementary matrices are diagonal which, in particular, are upper triangular.
Next, we can eliminate all entries above the diagonal (namely the entries u_{ij}

with $1 \leq i < j \leq n$ that are non-zero) by performing the following type III operations:

$$\begin{aligned}
&R_i \to -u_{in}R_n + R_i, \quad 1 \leq i \leq n-1,\\
&R_i \to -u_{i,n-1}R_{n-1} + R_i, \quad 1 \leq i \leq n-2,\\
&\quad \vdots\\
&R_i \to -u_{i3}R_3 + R_i, \quad 1 \leq i \leq 2,\\
&R_1 \to -u_{12}R_2 + R_1.
\end{aligned}$$

The corresponding type III elementary matrices are all upper triangular since the row operations are all of the form $R_i \to -u_{ij}R_j + R_i$ with $i < j$ strictly. The row operations above reduce U to I_n (hence reduce $[U|I_n]$ to $[I_n|V]$), which shows that U is invertible and $U^{-1} = V$ is the product of all the elementary matrices involved which are all upper triangular. Since products of upper triangular matrices are upper triangular, this proves that U^{-1} is an upper triangular matrix.

Question 10. Show that if A is a symmetric nonsingular matrix, then A^{-1} is also symmetric.

Solution

Since A is nonsingular,

$$AA^{-1} = A^{-1}A = I.$$

Taking transpose yields

$$(A^{-1})^T A^T = A^T (A^{-1})^T = I.$$

Since A is symmetric, it follows that

$$(A^{-1})^T A = A(A^{-1})^T = I.$$

The above identity means that $(A^{-1})^T$ is an inverse of A. But the uniqueness of inverse implies that

$$(A^{-1})^T = A^{-1}.$$

So, we conclude that A^{-1} is symmetric.

Question 11. Perform each of the following block multiplications:

(a) $\left[\begin{array}{ccc|c} 1 & 1 & 1 & -1 \\ 2 & 1 & 2 & -1 \end{array}\right] \left[\begin{array}{ccc} 4 & -2 & 1 \\ 2 & 3 & 1 \\ 1 & 1 & 2 \\ \hline 1 & 2 & 3 \end{array}\right]$

(b) $\left[\begin{array}{cc} 4 & -2 \\ 2 & 3 \\ 1 & 1 \\ \hline 1 & 2 \end{array}\right] \left[\begin{array}{ccc|c} 1 & 1 & 1 & -1 \\ 2 & 1 & 2 & -1 \end{array}\right]$

(c) $\left[\begin{array}{cc|cc} \frac{3}{5} & -\frac{4}{5} & 0 & 0 \\ \frac{4}{5} & \frac{3}{5} & 0 & 0 \\ \hline 0 & 0 & 1 & 0 \end{array}\right] \left[\begin{array}{cc|c} \frac{3}{5} & \frac{4}{5} & 0 \\ -\frac{4}{5} & \frac{3}{5} & 0 \\ \hline 0 & 0 & 1 \\ 0 & 0 & 0 \end{array}\right]$

(d) $\left[\begin{array}{ccc|cc} 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{array}\right] \left[\begin{array}{cc} 1 & -1 \\ 2 & -2 \\ 3 & -3 \\ \hline 4 & -4 \\ 5 & -5 \end{array}\right]$

Solution

(a) $\begin{bmatrix} 6 & 0 & 1 \\ 11 & -1 & 4 \end{bmatrix}$

(b) $\begin{bmatrix} 0 & 2 & 0 & -2 \\ 8 & 5 & 8 & -5 \\ 3 & 2 & 3 & -2 \\ 5 & 3 & 5 & -3 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(d) $\begin{bmatrix} 3 & -3 \\ 2 & -2 \\ 1 & -1 \\ 5 & -5 \\ 4 & -4 \end{bmatrix}$

Question 12. Find the inverse of each of the following matrices.

(a) $\begin{bmatrix} 1 & 2 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 2 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & 2 & 1 & 0 \\ 2 & 5 & 5 & 1 \\ -2 & -3 & 0 & 3 \\ 3 & 4 & -2 & -3 \end{bmatrix}$

Solution

(a) We perform row operations to reduce the augmented matrix:

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 2 & 0 & 0 & 1 \end{array}\right] \xrightarrow[R_3 \to -R_1 + R_3]{R_2 \to -R_1 + R_2} \left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & -2 & 0 & -1 & 1 & 0 \\ 0 & -1 & 1 & -1 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{R_3 \to -\frac{1}{2}R_2 + R_3} \left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & -2 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & -\frac{1}{2} & -\frac{1}{2} & 1 \end{array}\right]$$

$$\xrightarrow{R_2 \to -\frac{1}{2}R_2} \left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & \frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & 0 & 1 & -\frac{1}{2} & -\frac{1}{2} & 1 \end{array}\right]$$

$$\xrightarrow{R_1 \to -R_3 + R_1} \left[\begin{array}{ccc|ccc} 1 & 2 & 0 & \frac{3}{2} & \frac{1}{2} & -1 \\ 0 & 1 & 0 & \frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & 0 & 1 & -\frac{1}{2} & -\frac{1}{2} & 1 \end{array}\right]$$

$$\xrightarrow{R_1 \to -2R_2 + R_1} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{2} & \frac{3}{2} & -1 \\ 0 & 1 & 0 & \frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & 0 & 1 & -\frac{1}{2} & -\frac{1}{2} & 1 \end{array}\right]$$

Hence

$$\begin{bmatrix} 1 & 2 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 2 \end{bmatrix}^{-1} = \begin{bmatrix} \frac{1}{2} & \frac{3}{2} & -1 \\ \frac{1}{2} & -\frac{1}{2} & 0 \\ -\frac{1}{2} & -\frac{1}{2} & 1 \end{bmatrix}.$$

(b) We perform row operations to reduce the augmented matrix:

$$\left[\begin{array}{cccc|cccc} 1 & 2 & 1 & 0 & 1 & 0 & 0 & 0 \\ 2 & 5 & 5 & 1 & 0 & 1 & 0 & 0 \\ -2 & -3 & 0 & 3 & 0 & 0 & 1 & 0 \\ 3 & 4 & -2 & -3 & 0 & 0 & 0 & 1 \end{array}\right]$$

$$\xrightarrow[\substack{R_3 \to 2R_1 + R_3 \\ R_4 \to -3R_1 + R_4}]{R_2 \to -2R_1 + R_2} \left[\begin{array}{cccc|cccc} 1 & 2 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 3 & 1 & -2 & 1 & 0 & 0 \\ 0 & 1 & 2 & 3 & 2 & 0 & 1 & 0 \\ 0 & -2 & -5 & -3 & -3 & 0 & 0 & 1 \end{array}\right]$$

$$\xrightarrow[R_4 \to 2R_2 + R_4]{R_3 \to -R_2 + R_3} \left[\begin{array}{cccc|cccc} 1 & 2 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 3 & 1 & -2 & 1 & 0 & 0 \\ 0 & 0 & -1 & 2 & 4 & -1 & 1 & 0 \\ 0 & 0 & 1 & -1 & -7 & 2 & 0 & 1 \end{array}\right]$$

$$\xrightarrow[R_3 \to -R_3]{R_4 \to R_3 + R_4} \left[\begin{array}{cccc|cccc} 1 & 2 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 3 & 1 & -2 & 1 & 0 & 0 \\ 0 & 0 & 1 & -2 & -4 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 & -3 & 1 & 1 & 1 \end{array}\right]$$

$$\xrightarrow[R_3 \to 2R_4 + R_3]{R_2 \to -R_4 + R_2} \left[\begin{array}{cccc|cccc} 1 & 2 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 3 & 0 & 1 & 0 & -1 & -1 \\ 0 & 0 & 1 & 0 & -10 & 3 & 1 & 2 \\ 0 & 0 & 0 & 1 & -3 & 1 & 1 & 1 \end{array}\right]$$

$$\xrightarrow[R_2 \to -3R_3 + R_2]{R_1 \to -R_3 + R_1} \left[\begin{array}{cccc|cccc} 1 & 2 & 0 & 0 & 11 & -3 & -1 & -2 \\ 0 & 1 & 0 & 0 & 31 & -9 & -4 & -7 \\ 0 & 0 & 1 & 0 & -10 & 3 & 1 & 2 \\ 0 & 0 & 0 & 1 & -3 & 1 & 1 & 1 \end{array}\right]$$

$$\xrightarrow{R_1 \to -2R_2 + R_1} \left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & -51 & 15 & 7 & 12 \\ 0 & 1 & 0 & 0 & 31 & -9 & -4 & -7 \\ 0 & 0 & 1 & 0 & -10 & 3 & 1 & 2 \\ 0 & 0 & 0 & 1 & -3 & 1 & 1 & 1 \end{array}\right]$$

Hence

$$\begin{bmatrix} 1 & 2 & 1 & 0 \\ 2 & 5 & 5 & 1 \\ -2 & -3 & 0 & 3 \\ 3 & 4 & -2 & -3 \end{bmatrix}^{-1} = \begin{bmatrix} -51 & 15 & 7 & 12 \\ 31 & -9 & -4 & -7 \\ -10 & 3 & 1 & 2 \\ -3 & 1 & 1 & 1 \end{bmatrix}.$$

Question 13. Let

$$A = \begin{bmatrix} 2 & 1 \\ 6 & 5 \end{bmatrix}.$$

(a) Express A as a product of elementary matrices.

(b) Express A^{-1} as a product of elementary matrices.

Solution

(a) A can be reduced to the identity matrix by four row operations:

$$A = \begin{bmatrix} 2 & 1 \\ 6 & 5 \end{bmatrix} \xrightarrow{R_2 \to -3R_1 + R_2} \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$$

$$\xrightarrow{R_1 \to \frac{1}{2}R_1,\ R_2 \to \frac{1}{2}R_2} \begin{bmatrix} 1 & \frac{1}{2} \\ 0 & 1 \end{bmatrix}$$

$$\xrightarrow{R_1 \to -\frac{1}{2}R_2 + R_1} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

The corresponding elementary matrices are

$$E_1 = \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix}, \quad E_2 = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{bmatrix}, \quad E_3 = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{bmatrix}, \quad E_4 = \begin{bmatrix} 1 & -\frac{1}{2} \\ 0 & 1 \end{bmatrix}.$$

Therefore,

$$E_4 E_3 E_2 E_1 A = I$$

and hence

$$A = E_1^{-1} E_2^{-1} E_3^{-1} E_4^{-1} = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & \frac{1}{2} \\ 0 & 1 \end{bmatrix}.$$

(b) From part (a), we see that

$$A^{-1} = E_4 E_3 E_2 E_1 = \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 1 & -\frac{1}{2} \\ 0 & 1 \end{bmatrix}.$$

Question 14. Let

$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix},$$

where all four blocks are $n \times n$ matrices and A_{11} is invertible. Show that A can be factored into a product

$$\begin{bmatrix} I & O \\ B & I \end{bmatrix} \begin{bmatrix} A_{11} & A_{12} \\ O & C \end{bmatrix},$$

and determine B and C.

Solution

Note that

$$\begin{bmatrix} I & O \\ B & I \end{bmatrix} \begin{bmatrix} A_{11} & A_{12} \\ O & C \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} \\ BA_{11} & BA_{12} + C \end{bmatrix}.$$

The above will be equal to A if we set

$$\begin{cases} BA_{11} = A_{21}, \\ BA_{12} + C = A_{22}. \end{cases}$$

This implies that

$$\begin{cases} B = A_{21}A_{11}^{-1}, \\ C = A_{22} - BA_{12} = A_{22} - A_{21}A_{11}^{-1}A_{12}. \end{cases}$$

Therefore, A can be factored into the product

$$A = \begin{bmatrix} I & O \\ A_{21}A_{11}^{-1} & I \end{bmatrix} \begin{bmatrix} A_{11} & A_{12} \\ O & A_{22} - A_{21}A_{11}^{-1}A_{12} \end{bmatrix}.$$

Question 15. Let

$$P = \begin{bmatrix} A & O \\ B & C \end{bmatrix},$$

where A and C are invertible matrices. Find P^{-1} in terms of A^{-1}, C^{-1} and B.

Solution

Let

$$P^{-1} = \begin{bmatrix} W & X \\ Y & Z \end{bmatrix}.$$

Then

$$\begin{bmatrix} I & O \\ O & I \end{bmatrix} = PP^{-1} = \begin{bmatrix} AW & AX \\ BW + CY & CZ \end{bmatrix},$$

which implies that

$$\begin{cases} AW = I, \\ AX = O, \\ BW + CY = O, \\ CZ = I. \end{cases}$$

Hence, we deduce the following:

$$\begin{cases} W = A^{-1}, \\ X = O, \\ Y = -C^{-1}BW = -C^{-1}BA^{-1}, \\ Z = C^{-1}. \end{cases}$$

Therefore,

$$P^{-1} = \begin{bmatrix} A^{-1} & O \\ -C^{-1}BA^{-1} & C^{-1} \end{bmatrix}.$$